

Deep Research Report — ^1H -NMR Peak-List Classification

Project: DDLS 2025 Final Project — Alfred Larsson

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Agent: ChatGPT (GPT-5 Thinking)

1) Objective

Run a “Deep Research” to survey existing research, alternative approaches, and relevant data sources for classifying small molecules into metabolite families from **^1H -NMR peak-list spectra**. Provide concrete sources, data access points, and methodological options suitable for a 2–3 week project.

2) Key Findings at a Glance

- **Data sources:** HMDB (peak-list TXT, spectrum pages), BMRB (metabolomics entries, GISSMO library), nmrshiftdb2 (open database of assigned shifts & prediction), and COLMAR (query, ppm tools, HSQC databases).
 - **Methods used in literature:** Random Forest, SVM/PLS-DA/OPLS-DA baselines; CNNs and autoencoders for 1D spectra; newer **GNNs/transformers** for shift prediction and spectral learning.
 - **Augmentation:** **Predict chemical shifts** from structures (COLMARppm, PROSPRE, nmrshiftdb2) to expand training data and **normalize by solvent/field**.
 - **Feasibility:** HMDB + BMRB provide enough compounds to build a multi-class classifier across superclasses (Amino acids, Sugars, Lipids, Nucleotides, Organic acids) within the course constraints.
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3) Datasets & Access Points

3.1 Human Metabolome Database (HMDB)

- **Peak-list bundle:** “Raw NMR Spectra Peaklist Files (TXT)” (contains ppm/intensity lists).
- **Per-spectrum pages** provide *List of chemical shift values (TXT)* and **nmrML/JDX**.
- **License/terms:** Free to use with acknowledgement; commercial use requires permission.
- **Notes for this project:** HMDB TXT files carry HMDB Accessions; match to XML metadata for **Class/Superclass** labels.

Links:

- Downloads index (shows “Raw NMR Spectra Peaklist Files (TXT)”).

- Example spectrum pages with **TXT peak lists** in water at 500 MHz.

3.2 BMRB (Biological Magnetic Resonance Data Bank)

- **Metabolomics portal** with bulk downloads (entries tarball, relational CSV/SQL).
- **GISSMO** library provides curated small-molecule spectra; downloadable in **NMR-STAR** and **NMReDATA** formats.
- **Use in project:** Supplement HMDB with additional compounds; harmonize to **peak lists** via vendor tools or nmrML parsers if needed.

3.3 nmrshiftdb2

- Open database supporting **assigned shifts** and **prediction** ($^1\text{H}/^{13}\text{C}$).
- **Use in project:** Generate **predicted ^1H peak lists** (solvent/field aware if available) to **augment** minority classes or to pretrain an **autoencoder**.

3.4 COLMAR / COLMARppm

- Web tools and databases (incl. HSQC for mixtures) and **COLMARppm** for **rapid $^1\text{H}/^{13}\text{C}$ shift prediction** and automated assignment.
- **Use in project:** Benchmark or augment with predicted shifts; check ppm tolerances and solvent conditions.

4) Methods Landscape & Alternative Approaches

4.1 Supervised Baselines (Shallow)

- **RF / SVM / Logistic / PLS-DA** on binned ^1H spectra (e.g., 0.01–0.04 ppm bins).
- **Pros:** Fast, interpretable feature importances, aligns with your current pipeline.
- **Refs:** Evaluation studies comparing multivariate classifiers on simulated/experimental ^1H -NMR; themed reviews on ML in NMR.

4.2 Deep Learning for 1D ^1H Spectra

- **1D-CNN** for sequence-like spectra; **denoising autoencoders** $\rightarrow$ classifier.
- **Spectrum-aware models:** e.g., **NMRQNet** for identification/quantification from NMR spectra.
- **Transformers / contrastive:** pretrain on predicted/real spectra (SimCLR-style) to learn ppm-invariant embeddings.

4.3 Graph-Based Chemical-Shift Predictors (for Augmentation)

- **GNN-based** chemical-shift prediction from small data; **PROSPRE** (deep learning, solvent-aware ^1H shift prediction).
- **Augmentation idea:** From **SMILES** $\rightarrow$ predicted **^1H shifts** $\rightarrow$ synthesize peak lists (add line-shapes if needed), then train classifier; helpful for **class imbalance**.

4.4 Domain-Specific Tools

- **COLMARppm** for fast shift predictions and assignments (can generate clean synthetic peaks).
 - **GISSMO** curated lineshapes aid realistic synthetic spectra (Lorentzian/Gaussian).
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5) Practical Recommendations for This Project

1. **Core dataset:** HMDB peak-list TXT + matched **Superclass/Class** from HMDB XML (stream parse).
 2. **Supplement:** Add BMRB/GISSMO entries where classes are under-represented.
 3. **Preprocessing:**
 - **Solvent/field normalization:** keep water/D2O and 500 MHz subsets; otherwise apply **ppm alignment** and **intensity scaling** (unit max or total integral).
 - **Binning:** 0.02 ppm bins in [-1, 14] ppm is a good start; test 0.01 and 0.04.
 - **Peak to vector:** convert lists to binned intensities; optionally **peak-picking** → **sparse vector**.
 4. **Baselines:** RF, Logistic, and **PLS-DA** with 5-fold, report **balanced accuracy & weighted F1**.
 5. **Improvements:**
 - **1D-CNN** on normalized vectors; small architecture (100k params).
 - **Autoencoder pretraining** or **contrastive pretraining** (real+predicted spectra).
 - **Class weights** or **SMOTE-like augmentation** with **predicted shifts** for minority classes.
 6. **Error analysis:** Confusion matrix by **Superclass**; inspect ppm regions driving misclassifications (use permutation/SHAP on bins).
 7. **Accessibility:** Minimal **Streamlit** app: upload peak list → binned vector → prediction + top-bins explanation.
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6) Risks & Mitigations

- **Instrument/solvent heterogeneity:** Restrict to consistent conditions (e.g., H₂O/D₂O, 500 MHz) or apply alignment; document solvent/field in features.
 - **Class imbalance:** Use **class weights** and **shift-prediction augmentation** (COLMARppm/PROSPRE).
 - **XML size (HMDB):** **Streaming parse** to extract only accession + class fields.
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7) Citations & Source Pointers

- **HMDB Downloads** (lists “Raw NMR Spectra Peaklist Files (TXT)”): *Human Metabolome Database — Downloads*.

- Example HMDB spectrum pages with **TXT lists** and nmrML/JDX: examples for 500 MHz H¹ O.
- **BMRB metabolomics** bulk downloads and relational DB; **GISSMO** library page.
- **nmrshiftdb2** overview & usage; GNN shift-prediction on nmrshiftdb2 data (Rull 2023).
- **COLMARppm** (Rigel 2023/2024) — ACS Anal. Chem. article & PubMed; free PMC article.
- **NMRQNet** (Wang 2023) for deep NMR analysis.
- **Evaluation of multivariate classifiers** for 1D ¹H-NMR (Vu 2019).
- **Tea origin classification** via ¹H-NMR + ML (Hou 2024) — an applied example.
- Review: **Machine learning in NMR spectroscopy** (Klukowski 2025).
- Perspective on metabolomics & COLMAR databases (Markley 2017).

(Full URLs and metadata embedded in the accompanying transcript below.)

8) Search Transcript (Prompts & Results)

Prompt 1: “HMDB NMR peaklist files proton NMR peaks TXT download HMDB”

Result: HMDB downloads page lists “Raw NMR Spectra Peaklist Files (TXT)”.

Prompt 2: “classification of metabolites from 1H NMR spectra machine learning study”

Results: NMRQNet (2023); classifier comparisons for 1D ¹H-NMR (2019); applied studies (e.g., 2024 tea origin).

Prompt 3: “BMRB small molecule NMR database metabolomics peak lists download”

Results: BMRB metabolomics portal; bulk downloads; GISSMO library.

Prompt 4: “nmrshiftdb2 dataset download predicted 1H NMR chemical shifts small molecules”

Results: nmrshiftdb2 site, documentation; shift-prediction GNN paper (2023).

Prompt 5: “COLMAR NMR database small molecules 1H 13C metabolomics”

Results: COLMARppm articles (2023/2024) describing fast ¹H/¹³C shift prediction and assignment.

9) How to Cite in Your Plan

- Cite HMDB downloads and specific spectrum pages you used (include accession IDs).
- If using BMRB/GISSMO, cite the metabolomics portal and GISSMO.

- If augmenting with predicted shifts, cite **COLMARppm** and/or **PROSPRE** with solvent details.
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10) Appendix: Direct Links

- HMDB Downloads — “Raw NMR Spectra Peaklist Files (TXT)”.
 - HMDB per-spectrum examples (TXT lists, 500 MHz H¹ O).
 - BMRB metabolomics portal; bulk downloads.
 - GISSMO library.
 - nmrshiftdb2 home; usage guide; Rull 2023 GNN shift prediction.
 - COLMARppm (ACS Anal. Chem. paper; PubMed; PMC open text).
 - NMRQNet (PMC).
 - Classifier evaluation for 1D ¹H-NMR (PMC).
 - Machine learning in NMR spectroscopy (2025 review).
 - Metabolomics perspective incl. COLMAR (2017).
 - Applied example (tea origin classification, 2024).
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Prepared for inclusion as the “Deep Research transcript/report” in the DDLS Step-1 Project Plan.